Multi-Stakeholder Ethical Decision-Making Frameworks for AI Systems

CCDS25-0582 — Final Year Project Defence

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Presentation Outline

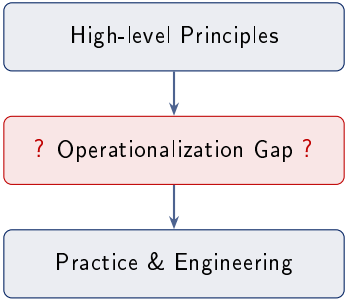
1. Background and Motivation
2. Framework Design
3. System Architecture
4. Algorithm Design
5. Platform Interface
6. Case Study Evaluation
7. Discussion
8. Limitations and Future Work

Background: The Urgent Need for AI Ethics

High-profile AI failures:

- Biased facial recognition with higher error rates for women and people of color (Buolamwini & Gebru, 2018)
- Automated hiring tools that perpetuate age and gender discrimination (EEOC v. iTutorGroup, 2022)
- Unauthorized use of 1.6 million patient records (NHS–DeepMind, 2017)

Key insight: Technical performance metrics alone are insufficient for evaluating AI acceptability.



Motivation: Three Interconnected Gaps

1. **Lack of standardized, quantifiable metrics.**
   Ethical concepts remain abstract and context-dependent; evaluations are subjective and hard to compare.

2. **Unmanaged stakeholder conflict.**
   Developers, regulators, and affected communities hold different values and risk tolerances. Traditional processes optimize for a single perspective.

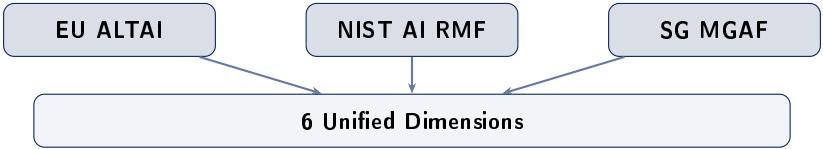
3. **Absence of auditable, reproducible workflows.**
   Ethical assessments are ad hoc and recorded in static documents, making reproduction and audit nearly impossible.

Project Goal

Design and implement a computational framework that enables organizations to **systematically and reproducibly** evaluate AI systems against multiple governance standards, while explicitly managing conflicting stakeholder priorities.

Governance Frameworks Integrated in MEDF

EU ALTAI	NIST AI RMF	SG MGAF
Rights-based approach	Risk-management process	Industry-focused guidance
7 requirements → 22 assessment questions	4 core functions → 19 subcategories	2 guiding principles → 20 principles
Strongest on:	*Strongest on:*	*Strongest on:*
Privacy, Human Oversight	Accountability, Safety	Transparency, Accountability



Six Unified Dimensions

1. Transparency & Explainability
2. Fairness & Non-discrimination
3. Safety & Robustness
4. Privacy & Data Governance
5. Human Agency & Oversight
6. Accountability

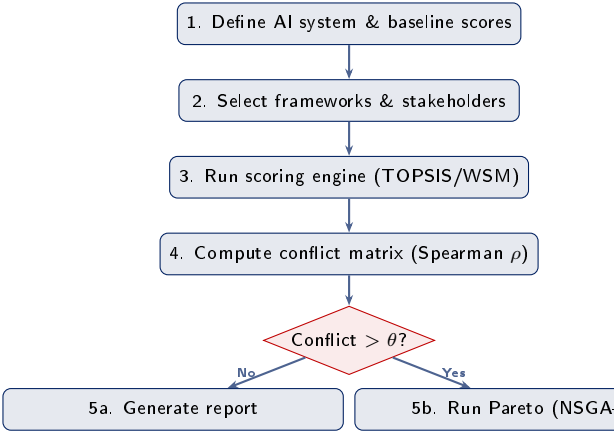
- Derived through systematic analysis of overlapping themes across all three frameworks.
- Each dimension scored on a 1–7 Likert scale, normalized to $[0, 1]$.
- Framework definitions stored as version-controlled YAML files, ensuring transparency and auditability.

Framework weights are derived from the structural emphasis of each governance document (number of requirements, clauses, or principles mapped to each dimension).

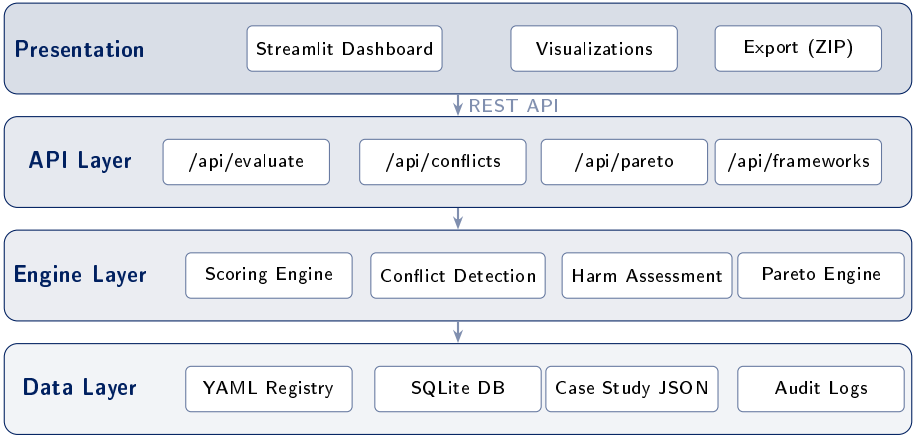
Core Design Principles and Workflow

Design Principles:

- **Modularity:** Frameworks, stakeholders, and algorithms are interchangeable components.
- **Reproducibility:** Same inputs always produce the same outputs.
- **Stakeholder-centricity:** Configurable weightings model diverse values.
- **Quantification:** Translates qualitative ethics into decision-support metrics.



Three-Tier System Architecture



Technology Stack

Backend:

- **FastAPI** – high-performance async API server with automatic OpenAPI documentation
- **Pydantic** – strict data validation for request/response schemas
- **SQLAlchemy** – ORM for SQLite persistence
- **NumPy** – numerical computation for scoring engines
- **pymoo** – NSGA-II multi-objective optimization

Frontend & Testing:

- **Streamlit** – interactive data-centric web dashboard
- **Plotly** – interactive charts and visualizations
- **Pytest** – 105 test functions across 33 test files (unit, integration, property-based, end-to-end)
- **GitHub Actions** – CI/CD on every push

Deployment:

- Backend: Render
- Frontend: Streamlit Cloud

`https://ccds25-0582-medf-api.streamlit.app/`

Scoring Engine: Effective Weight Computation

Step 1: Compute effective weights via product pooling of stakeholder and framework weights:

$$w_{\text{eff}} \propto w_s \odot w_f \tag{1}$$

where w_s = stakeholder weight vector (normalized), w_f = framework intrinsic weight vector (normalized).

Interpretation: A dimension receives high effective weight *only if both* the stakeholder *and* the framework consider it important. This prevents any single source from dominating.

Example

A *Developer* assigns Safety = 0.30. If ALTAI's intrinsic weight for Safety is also high (0.22), the effective weight is amplified. But if evaluating under MGAF (Safety = 0.16), the effective weight is moderated.

Scoring Engine: TOPSIS Algorithm

TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution):

1. **Normalize** Likert scores (1–7) to [0, 1]: $x'_{ij} = \frac{x_{ij} - 1}{7 - 1}$
2. **Vector normalize**: $r_{ij} = \frac{x'_{ij}}{\sqrt{\sum_i (x'_{ij})^2}}$
3. **Apply effective weights**: $v_{ij} = r_{ij} \times w_{\text{eff},j}$
4. **Determine** ideal (A^+) and anti-ideal (A^-) solutions
5. **Euclidean distances**: $d^+ = \sqrt{\sum_j (v_{ij} - A_j^+)^2}, \quad d^- = \sqrt{\sum_j (v_{ij} - A_j^-)^2}$
6. **Closeness coefficient**: $C = \frac{d^-}{d^+ + d^-}$

$C = 1.0 \rightarrow$ perfect match with ideal $\qquad C = 0.0 \rightarrow$ match with anti-ideal

Conflict Detection and Consensus Generation

Conflict Detection:

Spearman's rank correlation (ρ) between stakeholder contribution vectors:

$$\text{contribution}_k = w_{\text{eff},k} \odot s_{\text{norm}}$$

- $\rho \approx +1$: strong agreement
- $\rho \approx 0$: no relationship
- $\rho \approx -1$: strong disagreement

Contribution-based metric captures case-specific dynamics, not just general priority differences.

Consensus via NSGA-II:

Multi-objective optimization to find Pareto-optimal weight vectors:

$$f_k(w_c) = \sum_i s_i \cdot |w_{c,i} - w_{k,i}|$$

- Minimizes salience-weighted distance to *all* stakeholders simultaneously
- Result: Pareto frontier of viable compromises
- No stakeholder can improve without worsening another

Harm Assessment

Platform Interface: Evaluation View

Configuration Panel:

- Select AI system and governance frameworks
- Adjust baseline dimension scores via Likert sliders (1–7)
- Choose scoring method: TOPSIS or WSM
- Select stakeholder profiles with customizable weights

Results Display:

- Overall ethical score with risk classification
- Per-framework score breakdown
- Dimensional contribution radar chart

MEDF: Multi-Stakeholder Ethical Decision Framework

Overall Score: **0.6307** (Moderate)

Radar Chart: Dimensional Contributions

Per-Framework Scores Table

Conflict Level: **HIGH** (0.086)

Platform Interface: Conflict and Consensus Views

Conflict Detection:

- Heatmap of Spearman ρ between all stakeholder pairs
- Priority conflict (weights-only) and system-salience conflict (contribution-based)
- Identifies most divergent dimensions per pair

Dev	Res	Aff
1.00	0.09	−0.31
0.09	1.00	0.26
−0.31	0.26	1.00

Pareto Resolution:

- NSGA-II generates 25 non-dominated consensus solutions
- Each solution is a viable compromise weight vector
- Interactive selection to explore trade-offs
- Consensus radar chart shows recommended weights

Export:

- Full analysis bundle (ZIP) containing evaluation results, conflict matrices, and Pareto solutions
- Methodology summary for audit trail

Stakeholder Weight Profiles

Dimension	Developer	Regulator	Affected Community
Transparency	0.10	0.20	0.10
Fairness	0.15	0.20	0.30
Safety	0.30	0.10	0.10
Privacy	0.15	0.15	0.15
Human Oversight	0.15	0.10	0.20
Accountability	0.15	0.25	0.15
Total	1.00	1.00	1.00

- **Developer:** Emphasizes Safety (0.30) – building reliable systems.
- **Regulator:** Emphasizes Accountability (0.25), Transparency (0.20), Fairness (0.20).
- **Affected Community:** Emphasizes Fairness (0.30) and Human Oversight (0.20).

Case Study 1: Metropolitan Police Live Facial Recognition

System: Real-time face scanning in public spaces against watchlists.

Key issues:

- Higher error rates for women and people of color
- Mass surveillance concerns
- Chilling effect on right to protest

Dimension	Score
Transparency	2.5
Fairness	1.5
Safety	5.0
Privacy	1.5
Human Oversight	2.5
Accountability	4.0

MEDF Results (TOPSIS, all frameworks):

Framework	Dev	Reg	Aff
EU ALTAI	0.4583	0.3092	0.2462
NIST RMF	0.5552	0.4537	0.3823
SG MGAF	0.4344	0.3497	0.2905
All	0.4826	0.3709	0.3063

Overall: 0.3866 – Critical Risk

Clear divergence: Developer (0.48) vs. Affected Community (0.31). Contribution-based conflict: $\rho = 0.89$–0.94 (low conflict on *ranking*, divergent on *magnitude*).

Case Study 2: iTutorGroup Automated Hiring Screening

System: Automated applicant screening that rejected females 55+ and males 60+.

Key issues:

- Deliberate age and sex discrimination
- EEOC investigation → $365,000 settlement
- Systemic bias encoded as business rules

Dimension	Score
Transparency	2.5
Fairness	1.5
Safety	3.0
Privacy	3.0
Human Oversight	2.0
Accountability	4.5

MEDF Results (TOPSIS, all frameworks):

Framework	Dev	Reg	Aff
EU ALTAI	0.3185	0.3418	0.2462
NIST RMF	0.3787	0.4892	0.3785
SG MGAF	0.3292	0.3702	0.2766
All	0.3421	0.4004	0.3005

Overall: 0.3477 – Critical Risk

Catastrophic fairness failure renders the system unacceptable from *every* perspective. Moderate conflict: Dev–Reg $\rho = 0.54$, Dev–Aff $\rho = 0.77$.

Case Study 3: Royal Free NHS–DeepMind Streams App

System: Clinical decision-support app for acute kidney injury detection.

Key issues:

- Access to 1.6M patient records without explicit consent
- ICO ruled: breach of data protection law
- *But:* genuine clinical benefit and high safety

Dimension	Score
Transparency	4.0
Fairness	4.0
Safety	**6.0**
Privacy	3.5
Human Oversight	5.0

MEDF Results (TOPSIS, all frameworks):

Framework	Dev	Reg	Aff
EU ALTAI	0.6673	0.5376	0.5475
NIST RMF	0.7092	0.5836	0.5847
SG MGAF	0.6479	0.5470	0.5780
All	0.6748	0.5561	0.5701

Overall: 0.6003 – Medium Risk

Highest conflict: Dev–Reg $\rho = -0.14$, Reg–Aff $\rho = 0.03$. Balanced profile exposes *genuine* disagreements about priorities.

Cross-Case Comparative Analysis

Case Study	Overall	Risk	Key Weakness	Max Conflict (ρ)
Facial Recognition	0.3866	Critical	Fairness, Privacy	0.89 (Low)
iTutorGroup Hiring	0.3477	Critical	Fairness	0.54 (Moderate)
NHS–DeepMind Streams	0.6003	Medium	Privacy	−0.14 (High)

Key patterns:

1. Systems with severe ethical violations (Cases 1 and 2) are correctly flagged as Critical Risk.
2. Complex benefit-vs-harm trade-offs (Case 3) produce nuanced Medium Risk results.
3. **Contribution-based conflict varies across cases** even with *identical* stakeholder weights:
   - Extreme profiles → low conflict (everyone agrees on the ranking)
   - Balanced profiles → high conflict (genuine priority disagreements surface)
4. This validates MEDF's ability to distinguish genuinely contentious cases from clear violations.

Key Findings and Contributions

Critical Findings:

1. **Quantification clarifies debates.**
   Assigning numerical scores transforms vague ethical discussions into structured, actionable analysis.
2. **Trade-offs are made explicit.**
   Conflict analysis pinpoints *why* stakeholders disagree and which dimensions drive divergence.
3. **Multi-framework approach adds value.**
   Ethical scores vary by framework, reflecting jurisdictional priorities.

Position in the Landscape:

- Unlike *qualitative checklists*: MEDF provides computational, quantitative output.
- Unlike *fairness toolkits* (AI Fairness 360): operates at a higher abstraction level across all ethical dimensions.
- Unlike *governance platforms*: explicitly models multi-stakeholder conflict with algorithmic consensus.

Role: Decision-support tool that *augments* human judgment, not replaces it.

Limitations and Future Work

Limitations:

- **Input subjectivity:** Baseline scores and stakeholder weights are user-provided and may be biased.
- **Model simplification:** TOPSIS/WSM assume dimension independence; real ethical dimensions interact non-linearly.
- **Score reduction:** A single score cannot capture full ethical nuance.
- **No primary research:** Usability not yet validated with real users.
- **Technical debt:** Conflict analysis logic resides in router layer rather than a dedicated engine module.

Future Work:

- **Empirical validation:** User studies with practitioners, ethicists, and policymakers.
- **Automated score generation:** NLP techniques to derive scores from documentation and audit reports.
- **Longitudinal tracking:** Monitor ethical posture evolution over time.
- **CI/CD integration:** Embed ethical checks into development pipelines.
- **Expanded frameworks:** IEEE, OECD, and sector-specific regulations.

Conclusion

Summary of Contributions

1. **Operationalized ethical principles** into a quantitative, computational model by harmonizing three governance frameworks into six unified dimensions.
2. **Placed stakeholder conflict at the center** of ethical evaluation with configurable weights, Spearman conflict detection, and NSGA-II Pareto optimization.
3. **Delivered a functional, open-source research artifact** with comprehensive testing (105 tests), public deployment, and full documentation.

The responsible development of AI requires not only a commitment to ethical principles, but also practical tools to put those principles into action.

Thank you.

Questions?

Backup: Risk Classification Thresholds

TOPSIS Closeness Coefficient C:

Range	Risk Level
$C \geq 0.80$	Low
$0.60 \leq C < 0.80$	Medium
$0.40 \leq C < 0.60$	High
$C < 0.40$	Critical

Harm Score h:

Range	Severity
$h < 0.25$	Low
$0.25 \leq h < 0.50$	Moderate
$0.50 \leq h < 0.75$	High
$h \geq 0.75$	Critical

These two systems are conceptually consistent: C is "higher is better" while h is "higher is worse."

Backup: Framework Weight Distributions

Dimension	EU ALTAI	NIST RMF	SG MGAF
Transparency	0.12	0.12	0.22
Fairness	0.14	0.10	0.12
Safety	0.22	0.24	0.16
Privacy	0.20	0.12	0.12
Human Oversight	0.18	0.12	0.18
Accountability	0.14	0.30	0.20
Total	1.00	1.00	1.00

Weights derived from structural emphasis (number of requirements/subcategories/principles per dimension). Stored as version-controlled YAML files.

Backup: TOPSIS Detailed Computation Example

For facial recognition under EU ALTAI with Developer weights:

1. **Raw scores:** [2.5, 1.5, 5.0, 1.5, 2.5, 4.0]
2. **Normalized to [0,1]:** [0.25, 0.083, 0.667, 0.083, 0.25, 0.50]
3. **Stakeholder weights:** [0.10, 0.15, 0.30, 0.15, 0.15, 0.15]
4. **Framework weights:** [0.12, 0.14, 0.22, 0.20, 0.18, 0.14]
5. **Effective weights** (product pooled, re-normalized): emphasize Safety and Privacy
6. **TOPSIS result:** $C = 0.4583$

The low score is driven by the very low Fairness (0.083) and Privacy (0.083) scores, even though Safety is high (0.667).

Backup: API Endpoint Summary

Method	Endpoint	Description
GET	/api/health	Health check
GET	/api/frameworks	List governance frameworks
GET	/api/stakeholders	List stakeholder profiles
POST	/api/stakeholders	Create stakeholder profile
POST	/api/evaluate	Run ethical evaluation
POST	/api/conflicts	Compute conflict matrix
POST	/api/pareto	Run NSGA-II optimization

Full interactive documentation: `/api/openapi.json`
All schemas defined via Pydantic in `app/models.py`.